

Apoorva Kulkarni

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RESEARCH INTERESTS

Multimodal foundation models, with a focus on large audio and speech models: temporal reasoning, post-training, and evaluation.

EDUCATION

University of Maryland, College Park

M.S. in Computer Science, GPA: 3.96/4.0

Advised by Prof. Dinesh Manocha and Prof. Ramani Duraiswami (GAMMA Lab, PIRL Lab)

2025 – 2027

College Park, MD

PES University

B.Tech in Computer Science and Engineering, GPA: 9.69/10 (3.97/4.0)

Prof. CNR Rao Scholarship (top 2% of cohort in every semester)

2020 – 2024

Bangalore, India

PUBLICATIONS (* equal contribution)

[1] **TEMPO: Temporally-grounded Multi-task Post-training for Large Audio-Language Models**

Apoorva Kulkarni*, K. Jayakumar*, S. Ghosh, U. Aich, R. Duraiswami, D. Manocha. **EMNLP 2026** [project] [code]

[2] **A Closer Look at Failure Modes in Temporal Understanding of Large Audio-Language Models**

Apoorva Kulkarni, K. Jayakumar, S. Ghosh, S. Wiegrefe, D. Manocha, R. Duraiswami. **INTERSPEECH 2026** [project]

[3] **FESTA: Functionally Equivalent Sampling for Trust Assessment of Multimodal LLMs**

D. Bhattacharya*, Apoorva Kulkarni*, S. Ganapathy. **Findings of EMNLP 2025**

[4] **Benchmarking and Confidence Evaluation of Audio-LLMs for Temporal Reasoning**

D. Bhattacharya*, Apoorva Kulkarni*, S. Ganapathy. **INTERSPEECH 2025**

RESEARCH EXPERIENCE

Research Scientist Intern

June 2026 – August 2026

Hippocratic AI

- Curated a spoken-empathy benchmark of 215 counselor-client turn pairs from 133 motivational-interviewing sessions, preserving native client audio and expert counselor turns as human references.
- Evaluated empathic content and vocal delivery separately using two tracks over the same conversations: fixed client audio with model-generated responses for content, and fixed reference text with model synthesis for delivery.
- Built a two-stage LLM judge that annotates each response independently before pairwise comparison, suppressing the verbosity and warmth bias of direct pairwise judging (cross-judge κ 0.45 to 0.75).
- Showed that content and delivery rankings diverge: GPT Realtime 2 placed 7th of 8 on content but 2nd on delivery, while human counselors led delivery at 75.9% yet ranked only second on content.

Research Assistant

Sept 2025 – present

PIRL Lab and GAMMA Lab, University of Maryland, College Park

- TEMPO [1]:** built the first unified large audio-language model for timestamped generation, covering multi-speaker ASR, diarization, temporal grounding, dense captioning, and music captioning in one decoder.
- Introduced atomic timestamp tokens (0.1s), a time-aware projector injecting sinusoidal wall-clock encodings, and a distance-aware Gaussian loss; cut multi-speaker ASR WER from 69.7% to 43.5% and raised diarization mIoU from 44.4% to 71.1% over Qwen3-Omni. Post-trained a 7B model over a three-stage curriculum and applied GRPO with verifiable temporal rewards, a first for unified audio timestamping; released 119K training and 10K evaluation samples.
- Temporal reasoning interpretability [2]:** ran the first causal attention interventions for LALMs on a 1,657-question benchmark; redistributing attention across audio tokens beat increasing it (20.5% vs 15.8% fix rate), ruling out modality imbalance as the explanation, and layer-targeted scaling lifted accuracy from 55.9% to 59.1% without fine-tuning.

Research Assistant

Jan 2024 – Jun 2025

LEAP Lab, Indian Institute of Science (IISc)

- TREA [4]:** built a 600-question audio benchmark separating event ordering, duration, and counting, where the best large audio-language model scored under 50% against 80% for humans.
- Showed accuracy alone misranks these models, adding calibration error and a label-free uncertainty measure over semantically invariant perturbations: the most accurate system was the worst calibrated.
- FESTA [3]:** built a black-box, unsupervised uncertainty estimator probing a model with equivalent inputs, whose prediction should hold, and complementary inputs, whose prediction should flip, scoring uncertainty as KL divergence from an ideally consistent and sensitive model.
- Improved selective-prediction AUROC by 33% on visual and 30% on audio reasoning over the best baseline across seven multimodal LLMs, catching low-uncertainty hallucinations that entropy-based methods miss.

TEACHING & SERVICE

Teaching Assistant, DATA/MSML 602: Principles of Data Science (graduate), UMD

Fall 2026

Organizing team, [The Second DISPLACE Challenge](#) & Special Session, INTERSPEECH 2024

2024

TECHNICAL SKILLS

Machine Learning: pre-training, fine-tuning, post-training, reinforcement learning, interpretability, efficiency

Languages & Frameworks: Python, PyTorch, HuggingFace, NumPy, pandas, scikit-learn